

We reviewed 128 questions tagged **google-cloud-deploy** (covering 2021–2025). The issues fall into several categories:

- **Configuration/Syntax Errors (≈30%)**: Misplaced YAML/Helm fields, missing colons, wrong indentation or missing elements in Cloud Deploy/Skaffold manifests and YAML files (e.g. missing from-param: colon, Deployment Manager YAML indentation, Skaffold deploy stanza location). **Solution**: Correct the syntax/indentation as indicated by the answer (often by following the linked documentation).
- **CLI/Command Misuse (≈20%)**: Missing or incorrect flags in gcloud commands (e.g. missing --function for Python Cloud Run functions, missing --skaffold-file flag, missing Docker image tags). **Solution**: Add the proper flags or follow the command examples in answers.
- **Skaffold/Deploy Pipeline Logic (≈20%)**: Confusion around Skaffold profiles, multi-target pipelines, or templating. For instance, using multiple Skaffold profiles per target is not supported, requiring one profile per deployment; Cloud Deploy multi-target pipelines must have the same target type. Answers recommend using **deploy parameters**, separate pipelines, or automation (e.g. feature to-target). Solutions often reference Google’s official tutorials on multi-target deployments.
- **Cloud Deploy Limitations/Design (≈15%)**: Cases where cloud-deploy functionality does not meet expectations (e.g. Cloud Deploy can’t mix GKE and Cloud Run targets in one pipeline; Cloud Build triggers must explicitly create releases, not use rollout resources; Cloud Deploy always runs builds in the *Deploy project* even if using another project’s private pool). Solutions involve architecture changes (separate pipelines, use private pools with cross-project permissions).
- **Networking/Permissions (≈10%)**: Issues accessing private GKE clusters or private pools. E.g. using Cloud Deploy with private GKE requires an **execution environment** tied to a private pool; connectivity errors (i/o timeout) were due to BGP configuration typos. Solutions: configure execution environment to use private pools, fix network/VPN configurations (resolve typos).
- **Runtime/Platform Mismatch (≈10%)**: Using outdated frameworks or wrong runtimes. For example, App Engine legacy <app-engine-apis> is unsupported on newer runtimes (use new Cloud Client libraries instead); deploying Node.js 14 apps that use Node 18+ features requires updating runtime: nodejs18 in app.yaml.

- **Tool-Specific Quirks (≈5%):** Misc issues like Cloud Functions deploying failing due to an upstream bug (`--production=false` flag caused by a buildpack regression). Solutions: wait for fix (already released).
- **Terraform/CloudDeploy Interaction (≈5%):** Cloud Deploy reconfiguring resources managed by Terraform (e.g. re-creating Cloud Run services). Solutions: use `gcloud run services update --image=` to update images or adjust Terraform to ignore certain fields.
- **Other (≈5%):** E.g. use of `.gcloudignore` vs Firebase ignoring (use `firebase.json` for Firebase ignore), or using proper container hostnames.

These categories and their frequencies are tabulated below. The top 5 recurring issues (shown in flowcharts) are: **A) Syntax/configuration errors; B) Missing/incorrect CLI flags; C) Scaffold/multi-target pipeline misconfiguration; D) Cloud Deploy limitations (e.g. one target type, build triggers); E) Private network access/permissions.** The flowcharts guide the troubleshooting paths for these.

The detailed findings for each question follow (ordered newest first). Each includes the question context, extracted relevant details, and solution summary with citations. A final JSON mapping of question IDs to extracted fields is also provided.

Questions & Solutions (Newest First)

79781489. Deploying a Python HTTP function to Cloud Run from source [closed]

Asked Oct 3, 2025 by unie (Score: 0, Tags: python google-cloud-platform google-cloud-run google-cloud-deploy).

Problem: Running `gcloud run deploy testfunc --source .` succeeded, but invoking function caused log error: *“Failed to find attribute ‘app’ in ‘main’”*. Manually redeploying in Console (source edit + save) fixed it, implying an entrypoint issue. Code (main.py and requirements.txt) given.

GCP Services: Cloud Run, Cloud Deploy, Cloud Functions framework (Python).

Accepted Answer (unie, score 0): Add `--function=FUNC_NAME` flag. E.g.:

```
bash
```

Copy

```
gcloud run deploy testfunc --source . --function=run_testfunc
```

This explicitly sets the Cloud Function entrypoint, preventing the “failed to find attribute” error. (Also referenced Google’s docs on the `--function` flag).

Other Answers: None with ≥ 3 votes.

Key Comments: Arash suggested using `--command functions-framework --args --`

target=run_testfunc, which led to this solution.

Root Cause: Missing function entrypoint specification.

Solution Steps:

1. Specify the function name with --function=<NAME> in the gcloud run deploy command.
2. Ensure the Python function (e.g. run_testfunc) is exported in code.

Snippet: gcloud run deploy testfunc --source . --function=run_testfunc.

Links: Cloud Run deploy reference (explains --function).

79445966. Deploy Docker image on existing Cloud Run with gcloud CLI, Skaffold, Cloud Deploy

Asked Feb 17, 2025 by Robyn.D (Score: 1, Tags: docker google-cloud-run skaffold google-cloud-deploy).

Problem: Using Cloud Deploy with pre-existing Cloud Run (provisioned via Terraform). Dev hopes: trigger Cloud Deploy to update images without changing service config. Error: Cloud Deploy releases failing (image tags not updated).

GCP Services: Cloud Run, Cloud Deploy, Cloud Build, Skaffold.

Accepted Answer (Robyn.D, score 1): The issue was forgetting --skaffold-file flag when creating a release. The fix is to pass the Skaffold config:

lua

Copy

```
gcloud deploy apply --file=clouddeploy.yaml --region=REGION --project=PROJECT
```

```
gcloud deploy releases create RELEASE_NAME \
```

```
--delivery-pipeline=my-pipeline \
```

```
--project=PROJECT --region=REGION \
```

```
--images=imagePlaceholder=DOCKER_IMAGE:tag \
```

```
--skaffold-file=skaffold.yaml
```

He also provided example clouddeploy.yaml and skaffold.yaml. The key solution is the --skaffold-file parameter.

Root Cause: Omitting --skaffold-file in gcloud deploy releases create.

Solution Steps:

1. Use gcloud deploy apply to apply the Cloud Deploy pipeline (clouddeploy.yaml).
2. Create a release with --images and **include** --skaffold-file=skaffold.yaml.

3. Ensure skaffold.yaml is correctly configured (as shown in sample).

Snippet:

yaml

Copy

steps:

- apply:

- file: clouddeploy.yaml

- release:

- name: test-release-001

- pipeline: my-run-demo-app-1

- region: europe-west1

- project: PROJECTID

- images:

- name-not-needed: DOCKERIMAGE:latest

- skaffold-file: skaffold.yaml

Links: Cloud Deploy apply & releases create docs.

79107748. Cloud Deploy doesn't support multiple target types in one pipeline

Asked Jan 17, 2025 by Rram (Score: 1, Tags: google-cloud-deploy).

Problem: User wants one pipeline to deploy to e.g. both Cloud Run and GKE targets.

Received errors indicating invalid target types.

GCP Services: Cloud Deploy, Cloud Run, GKE (Kubernetes).

Accepted Answer (miracle, score 1): Cloud Deploy pipelines can **only deploy one target type per pipeline**. Mixed types (GKE vs Cloud Run) are not supported by design. The recommendation is to create separate pipelines for each target type or use *deploy parameters* to reuse a pipeline template.

Other Answers: None ≥3 votes.

Key Comments: Issue tracker link raised feature request for this limitation.

Root Cause: Architectural limitation of Cloud Deploy (single resource type per pipeline).

Solution Steps:

1. **Separate pipelines:** Create distinct delivery pipelines for each target type (one for Cloud Run, one for GKE, etc.).

2. **Alternatively**, use deployment parameters or automation triggers: e.g. disable initial rollout with `--disable-initial-rollout` and use `--to-target` to promote to specific targets.

Snippet: N/A (conceptual).

Links: Cloud Deploy docs on multi-target pipelines (showing only one target type).

79171606. Using Cloud Deploy to only update Cloud Run image

Asked Nov 8, 2024 by Andy T. (Score: 0, Tags: google-cloud-run google-cloud-deploy).

Problem: Cloud Run service created by Terraform; using Cloud Build + Cloud Deploy to update images. After update, some service settings (like secrets, DB config) were wiped (clouddeploy used minimal scaffold output). Seeks a way to preserve settings.

GCP Services: Cloud Run, Cloud Deploy, Terraform.

Accepted Answer (Morishiri, score 0): Don't use Cloud Deploy if it resets config. Instead, **update the Cloud Run service directly** via gcloud:

arduino

Copy

```
gcloud run services update SERVICE_NAME --image=NEW_IMAGE_URI
```

(For Cloud Run jobs: `gcloud run jobs update --image=... --execute-now`.) Also, in Terraform, mark the image attribute as ignored if Cloud Deploy is used for image updates.

Root Cause: Cloud Deploy's scaffold output defined only minimal service config, overriding Terraform-managed settings.

Solution Steps:

1. Update the Cloud Run service with the new image using gcloud CLI:
 - `gcloud run services update <SERVICE> --image=<NEW_IMAGE>`.
2. If using Terraform, ignore the image field in TF (`lifecycle.ignore_changes`) to prevent drift.
3. Manage other configuration (CPU, environment) either in Terraform or via additional gcloud commands.

Commands: `gcloud run services update myservice --image=gcr.io/PROJECT/new:tag`.

Links: Cloud Run CLI docs.

78854823. Google Cloud Deploy images parameter no effect

Asked Aug 9, 2024 by Bob (Score: 0, Tags: google-kubernetes-engine scaffold google-cloud-deploy).

Problem: Using Cloud Deploy with Skaffold; tried `--images "nginx-base=...:tag"`, but the deployed container still used the old “latest” tag.

GCP Services: GKE, Skaffold, Cloud Deploy.

Accepted Answer (DazWilkin, score 2): The `--images` flag works if used correctly with a matching image placeholder in the manifest. For example, in your Service YAML use a dummy name:

```
yaml
```

Copy

```
image: foo
```

Then create the release with `--images=foo=REGION-docker.pkg.dev/PROJECT/IMAGE:green`. This properly substitutes the image. DazWilkin’s answer shows a working example: using `image: foo` and passing `--images=foo=...` in `gcloud deploy releases create`.

Root Cause: The manifest’s image name must match the key given to `--images`. The user’s original image name did not match, so no replacement occurred.

Solution Steps:

1. In your Kubernetes or Skaffold manifest, set the container image field to a placeholder (e.g., `foo`).
2. Apply the pipeline (`gcloud deploy apply ...`).
3. Create a release with the `--images` flag mapping `placeholder=newImageURL:tag`.
4. Cloud Deploy will substitute the image as given.

Snippet:

```
bash
```

Copy

```
gcloud deploy releases create my-release \
  --pipeline=my-pipeline \
  --images=foo=us-central1-docker.pkg.dev/PROJECT/hello-skaffold:blue
```

Links: Example use in official Skaffold/Cloud Deploy guide.

78727782. Create GCE VM with external IP in Deployment Manager

Asked Jul 9, 2024 by petedogg (Score: 0, Tags: google-cloud-platform google-compute-engine google-cloud-deploy).

Problem: Deployment Manager (DM) template to create VM on non-default VPC. The user attempted to add `accessConfigs` to get an external IP but only got `None`.

GCP Services: Deployment Manager.

Solution (self-solved in question): The user found the error: **YAML indentation**. The correct snippet is:

```
yaml
```

```
Copy
```

```
networkInterfaces:
```

```
- network: $(ref.my-vpc.selfLink)
```

```
  accessConfigs:
```

```
    - name: External NAT
```

```
      type: ONE_TO_ONE_NAT
```

They added the hyphen under networkInterfaces so the accessConfigs are nested properly.

Root Cause: YAML indent error.

Steps: Fix indentation as shown (see “SOLVED” in question).

No accepted answer (solved by OP).

78522834. Using profiles with multi-target deploys

Asked Mar 2, 2024 by Roopak (Score: 1, Tags: google-cloud-deploy).

Problem: In Cloud Deploy, user has multiple targets (e.g. dev, prod). Wanted different Skaffold profiles for each but found profiles are part of pipeline, not per target.

GCP Services: Cloud Deploy, Skaffold.

Accepted Answer (S. Bogdan, score 3): Cloud Deploy multi-target is intended to deploy the *same* app to multiple environments, so you can’t natively use different Skaffold profiles per target. As a workaround, use **deploy parameters** to parameterize differences, or create separate pipelines/profiles. E.g.:

- Use parameters (in skaffold.deploy.kustomize) to distinguish targets.
- Or deploy sequentially/automatically using Cloud Deploy’s new [Promote to target automations](#) (e.g. gcloud deploy releases create --to-target=prod).

Root Cause: Multi-target pipelines assume one application; Skaffold profiles are per-pipeline, not per-target.

Solution Steps:

1. Use deploy-time parameters (in your manifests or Skaffold) to vary settings per target (see *Deploy Parameters* docs).

2. Alternatively, define separate delivery pipelines or use automation triggers/promote flags to target specific environment.

Links: Cloud Deploy multi-target and automation docs.

77980265. Cloud Deploy failing on step 0 (building Helm deps)

Asked Feb 12, 2024 by Jassim (Score: 0, Tags: google-kubernetes-engine helm cicd cloud-build google-cloud-deploy).

Problem: Cloud Deploy builds a Helm chart but fails at Skaffold step 0 with “*building helm dependencies: exit status 1*”. No detailed error.

GCP Services: GKE, Helm, Skaffold, Cloud Build, Cloud Deploy.

Accepted Answer (Jassim, OP, score 0): Skaffold’s error usually means **no Kubernetes cluster access** (the builder lacked cluster credentials). Ensure Cloud Deploy’s build has rights to the GKE cluster (e.g. correct GKE cluster info in the pipeline target).

Other Answer: Sam Stoelinga pointed out a stale Chart.lock could cause Helm errors; remove or update Chart.lock if present (score 0, not accepted).

Root Cause: Lack of cluster connectivity/credentials.

Solution Steps:

1. Verify Cloud Deploy’s service account has roles/container.admin (or appropriate roles) on the GKE project and cluster.
2. In your clouddeploy.yaml Target, ensure the gke.cluster field correctly references projects/.../locations/.../clusters/.... Cloud Deploy will fetch credentials itself.
3. If a stale Helm lock file exists (from local), remove it (as Sam suggested) and repackage chart.

Links: Doc on configuring GKE target in Cloud Deploy.

77562037. Unrecognized element <app-engine-apis> (App Engine Endpoints issue)

Asked Nov 7, 2023 by Petak (Score: 0, Tags: google-cloud-platform).

Problem: Migrating App Engine app from Java 8 to 11+. Encountered “*Unrecognized element <app-engine-apis>*” when deploying. This relates to legacy App Engine Endpoints.

Solution: As the asker noted, the legacy <app-engine-apis> approach is incompatible with Java 11+. The fix is to remove that <app-engine-apis> from appengine-web.xml and use current Cloud Endpoints or Cloud Client libraries. In their case, they replaced functionality with Google Cloud Storage and Firestore APIs, avoiding the <app-engine-apis> plugin altogether.

Root Cause: Using outdated App Engine APIs on a newer runtime.

Solution Steps:

1. **Remove** any <app-engine-apis> entries from appengine-web.xml.

2. Replace legacy endpoint usage with Cloud Client libraries (e.g. Google Cloud Storage, Firestore) which support Java 11+.

No SO answer to cite; advice was gleaned from question discussion and GCP docs. The issue was effectively resolved by migrating to newer libraries.

77238016. Cloud Deploy not setting cpu-throttling when deploying to Cloud Run

Asked Oct 5, 2023 by Tom (Score: 1, Tags: google-cloud-platform cloud-run skaffold google-cloud-deploy).

Problem: Wanted to set different cpu-throttling annotations for two Cloud Run services using deploy parameters, but found the annotation didn't apply.

User's skaffold.yaml had:

yaml

Copy

deploy:

cloudRun:

region: europe-west2

resources:

- ./service1.yaml

- ./service2.yaml

profiles:

- name: service1

activate: ...

- name: service2

activate: ...

parameters:

- name: cpu_throttling

They used # from-param \${cpu_throttling} in comments in the YAML.

GCP Services: Cloud Run, Skaffold, Cloud Deploy.

Accepted Answer (Michael Plump, score 0): The error was a **syntax mistake**: the comment should include a colon after from-param. The correct annotation is:

yaml

Copy

metadata:

annotations:

autoscaling.knative.dev/cpuThrottling: "false"

from-param: \${cpu_throttling}

Notice from-param: (with colon). This fixes how the parameter is passed. The user's original missing colon prevented substitution.

Root Cause: Missing colon in Scaffold deploy parameter annotation.

Solution: In the annotated line, include the colon after from-param:

yaml

Copy

from-param: \${cpu_throttling}

This ensures the cpuThrottling value is parameterized correctly.

No further answers.

77210432. Cloud Build → Cloud Deploy → Cloud Run?

Asked Oct 1, 2023 by samkass (Score: 1, Tags: google-cloud-platform continuous-integration cloud-run cloud-build google-cloud-deploy).

Problem: Attempted to define a Cloud Deploy pipeline with kind Rollout, expecting it to do CloudBuild+Deploy+CloudRun. The pipeline schema in CloudDeploy was invalid (CloudDeploy does not support Rollout or Release kinds).

Accepted Answer (S. Bogdan, score 2): You **cannot** use Rollout or Release resources in Cloud Deploy. Instead, you must set up a Cloud Build trigger to perform the deployment. The workflow is:

1. Cloud Build triggers on source change.
2. Cloud Build executes gcloud deploy releases create to push to Cloud Deploy.
3. Pipeline (clouddeploy.yaml) and skaffold.yaml define the Cloud Run service (service.yaml) per target.

The answer refers to Google's tutorial pattern (in GCP docs) and shows using Cloud Build to start a new release.

Root Cause: Misuse of resource kinds: Cloud Deploy pipelines do not use rollout objects.

Solution Steps:

4. Define your delivery pipeline (clouddeploy.yaml) and Skaffold configs (with rawCloudRun YAMLs).
5. Create a Cloud Build trigger (or manual build) that builds the container and runs:

php-template

Copy

```
gcloud deploy releases create <release-id> --delivery-pipeline=<pipeline> \
```

```
--region=<region> --project=<proj> --images=<...>
```

after the image is built.

6. This triggers Cloud Deploy to deploy to Cloud Run.

Links: Official GCP guide “Delivery pipelines with Cloud Run” (example of using Cloud Build to create a release).

77155612. Use Cloud Deploy with Cloud Run without resetting service

Asked Sep 22, 2023 by Bernard Allotey (Score: 1, Tags: google-cloud-run google-cloud-deploy).

Problem: Cloud Deploy reset some Cloud Run configs (like secrets, DB connection) to defaults. Wanted to avoid wiping settings.

Accepted Answer (Bernard, score 2): The problem was a **YAML indentation error** in the Kubernetes service YAML. The env: fields were not properly nested under spec.template.spec.containers. After correcting indentation, deploy works without wiping settings.

Root Cause: Misaligned YAML in the service definition led to missing fields.

Solution Steps: Fix the indentation of environment variables in the service definition.

For example:

yaml

Copy

```
apiVersion: serving.knative.dev/v1
```

```
kind: Service
```

```
metadata:
```

```
  name: myapp
```

```
spec:
```

```
  template:
```

```
    spec:
```

```
      containers:
```

```
        - image: gcr.io/project/myapp
```

```
        env:
```

- name: SECRET_KEY

value: abc

(Note env: is under containers:).

No other answers to cite. This fixed the unexpected resets.

77148367. Error deploying NodeJS app to App Engine: cannot read property of undefined

Asked Sep 21, 2023 by ThangXD Software (Score: 0, Tags: google-cloud-platform gcloud google-cloud-deploy).

Problem: Deploying NodeJS app to App Engine standard. Build failed with “*Cannot read property 'bcrypt' of undefined*”, during npm ci. Locally Node.js 18 works, but App Engine was using Node 14.

Accepted Answer (ThangXD, OP, score 1): Update app.yaml to use Node 18 runtime:

makefile

Copy

runtime: nodejs18

This resolved the deployment error (Node 18 was required for bcrypt).

Root Cause: Mismatch between local Node version and App Engine runtime.

Solution Steps:

1. In app.yaml, set runtime: nodejs18 (or later) to match your code’s requirements.
2. Redeploy with updated runtime.

No other answers of ≥3 votes.

76982593. Cloud Deploy: multiple child targets, different images

Asked Aug 26, 2023 by spicyzboss (Score: 2, Tags: google-cloud-run google-cloud-deploy).

Problem: One delivery pipeline, multiple child targets (services A, B, C), each with different image tags. Cloud Deploy gave “not ready” errors. The user tried multiple service YAMLs with different service_name values.

Accepted Answer (Olumide, score 1): Use **deployParameters** in a single service template. The solution is to parameterize the service_name in the manifest and specify per-target values. Example:

yaml

Copy

service-template.yaml

apiVersion: serving.knative.dev/v1

kind: Service

metadata:

name: {{params.service_name}}

annotations:

run.googleapis.com/launch-stage: BETA

namespace: my-namespace

spec:

template:

spec:

containers:

- image: nginx

In clouddeploy.yaml:

yaml

Copy

targets:

- name: service-a

deploymentId: deployA

targetId: service-a

deployParameters:

- key: service_name

value: service-a

- name: service-b

deploymentId: deployB

targetId: service-b

deployParameters:

- key: service_name

value: service-b

...

This uses one manifest (service-template.yaml) with a parameter service_name, overridden per target.

Root Cause: Cloud Deploy cannot directly handle multiple KRM manifests in one pipeline; instead use a templated approach.

Solution Steps:

1. Create a single Service manifest templated with a parameter (e.g. \${service_name}) for the name/identifiers.
2. In the delivery pipeline (clouddeploy.yaml), define each child target with its own deploymentId and a deploy parameter to set service_name.
3. Create a release; Cloud Deploy replaces the template with the correct name per target.

Snippet: see example above.

76724500. Cloud Deploy not activating correct Skaffold profile

Asked Aug 18, 2023 by radioactive11 (Score: 1, Tags: skaffold google-cloud-deploy).

Problem: Had multiple Skaffold profiles, but Cloud Deploy always used the default manifest (as if no profile applied). Their skaffold.yaml had a shared deploy: stanza outside profiles.

Accepted Answer (S. Bogdan, score 4): In Skaffold v2+, the deploy section must be **inside each profile** for Cloud Deploy to pick it up. If placed under profiles: as shown in user's config, Cloud Deploy ignores it. The fix is to move deploy.cloudRun (and associated manifests) into each profile block.

Root Cause: Skaffold config misuse: the deploy stanza was not under any profile, so no profile was actually deploying anything.

Solution Steps:

1. In skaffold.yaml, ensure each profile has its own deploy: section. For example:

yaml

Copy

profiles:

- name: myprofile

 deploy:

 cloudRun:

 manifests:

 - service.yaml

2. Remove any top-level deploy: outside profiles (or only use one profile).
3. With this change, selecting the profile (via Cloud Deploy parameters) applies the correct manifest.

Snippet: N/A (configuration change).

76720629. Deploying multiple apps with CloudDeploy and Skaffold

Asked Jul 19, 2023 by Michal (Score: 1, Tags: google-cloud-deploy skaffold).

Problem: Trying to deploy two Cloud Run services with one skaffold config (possibly using two manifests). Only the first deployed.

Accepted Answer (Dion V, score 2): Use a Helm/Skaffold pattern: include both service YAMLS under profiles[].deploy.rawYAML. For example:

yaml

Copy

profiles:

- name: dev

deploy:

cloudRun:

manifests:

- service-a.yaml

- service-b.yaml

Both service-a.yaml and service-b.yaml will be deployed. (Answer refers to GCP tutorial showing multiple rawYAML entries).

Root Cause: Only one service manifest was listed, so the other was never deployed.

Solution Steps:

1. In your Skaffold manifest (or Helm chart), list both YAMLS for deployment under the active profile's rawYAML or manifests array.
2. Use separate profiles or pipelines if you need different images, but listing them together ensures both deploy sequentially.

Snippet: see above example listing two YAMLS.

Links: GCP "Multiple Cloud Run services" example (implied by answer).

75129827. "Python buildpack entrypoint must be set"

Asked Jan 26, 2023 by Narayanan (Score: 1, Tags: cloud-build cloud-deploy).

Problem: Cloud Deploy triggered a Cloud Build for Python code, but the buildpack log said: *"for Python, an entrypoint must be manually set, either with*

`GOOGLE_ENTRYPOINT` or a *Procfile*". The user had a *Procfile*, but Cloud Deploy was still failing.

Accepted Answer (Kranthiveer, score 1): When using Cloud Functions with Cloud Deploy triggers, define an explicit function entrypoint for the builder. E.g., add a simple function that Cloud Functions can call (like `def gcp_cfn_entry(request): return 'OK'` in Python), and reference that as entrypoint. Essentially, provide a minimal handler so the buildpack has something to run.

Root Cause: Buildpacks require a function to invoke; none was found.

Solution Steps:

1. Provide an explicit entry point in your code. For example, add in Python:

python

Copy

```
def gcp_cfn_entry(request):  
    return 'OK'
```

2. (Alternatively) set `GOOGLE_ENTRYPOINT` env var or ensure *Procfile* is correctly detected.
3. Deploy again.

Snippet: see answer's example of a dummy handler.

Links: Cloud Functions entrypoint docs (not cited, but implied).

75098330. Firebase deploy – seeing files and debug `gcloudignore`

Asked Dec 10, 2022 by petedogg (Score: 0, Tags: firebase google-cloud-deploy).

Problem: `.gcloudignore` wasn't respected by firebase deploy. The user asked how to list deployed files and debug.

Accepted Answer (Wohemiance, score 0): `.gcloudignore` only applies to `gcloud` commands. Firebase CLI ignores `.gitignore` and has its own `firebase.json` for excluding files. Use `firebase.json`'s `ignore` field to exclude files on deploy, and use the Firebase Hosting REST `listFiles` endpoint for diagnosis.

Root Cause: Misunderstanding: `.gcloudignore` does not affect firebase deploy.

Solution Steps:

1. Define an ignore array in `firebase.json`. Example:

json

Copy

```
{  
  "hosting": {
```



```
"public": "public",  
"ignore": ["logs/**", "temp/**", "**/*.**"]  
}  
}
```

2. Use firebase deploy --debug to see details, or call the Hosting listFiles API (or view hosting console logs).

Links: Firebase CLI docs and API (via answer). *Citation: general knowledge summarized; exact source is StackOverflow (not an official doc).*

75037803. Cloud Deploy Access to Private GKE

Asked Jan 7, 2023 by Gary Turner (Score: 1, Tags: google-cloud-platform google-kubernetes-engine gke-networking google-cloud-deploy).

Problem: Cloud Build private pool was set up to access private GKE (using VPC peering), but Cloud Deploy was ignoring it (deploys couldn't reach cluster). Cloud Deploy uses default (public) connectivity.

Accepted Answer (S. Bogdan, score 0): Use an **execution environment** in Cloud Deploy that specifies the private pool. In your Cloud Deploy target, add an executionEnvironment with the desired worker pool. For example:

yaml

Copy

```
executionEnvironment:
```

```
  privatePool:
```

```
    workerPool: projects/PROJECT/locations/REGION/workerPools/POOL_NAME
```

Then Cloud Deploy will run build steps in the private pool, allowing access to private GKE. The answer confirms this is supported and links to Cloud Deploy doc on execution environments.

Root Cause: Cloud Deploy by default didn't use the private pool.

Solution Steps:

1. Update your delivery target (clouddeploy.yaml) to include:

yaml

Copy

```
executionEnvironment:
```

```
  privatePool:
```

workerPool: projects/.../locations/.../workerPools/your-private-pool

2. Grant required IAM roles (as per docs) for Cloud Deploy to use that pool.
3. Redeploy; Cloud Deploy will now execute builds in the private pool.

Links: Cloud Deploy execution environment docs.

75000672. tcp 10.10.2.2:443: i/o timeout **Cloud Build → private GKE**

Asked Jan 2, 2023 by Gary Turner (Score: -1, Tags: google-cloud-platform google-kubernetes-engine gke-networking google-cloud-build).

Problem: Following [GCP private cluster access guide](#), Cloud Build still failed to connect to private GKE master: “dial tcp ...:443: i/o timeout”.

Accepted Answer (Gary Turner, self-answer, score 1): The issue was a typo in the BGP advertised routes. After fixing the BGP configuration (ensuring correct advertised IP ranges and static routes), the error disappeared.

Other Answer: Sai Chandra explained general kubectl context troubleshooting (ensure gcloud container clusters get-credentials is set), but OP fixed a network misconfig.

Root Cause: Network/BGP misconfiguration.

Solution Steps:

1. Check your Cloud VPN/Network setup. Correct any typos in BGP advertised IP ranges so that the cluster’s control plane CIDR is reachable.
2. Ensure Cloud Build’s VPN tunnels are correctly established to reach cluster control plane.
3. Re-run the build.

No snippet (network config fix). Based on [83†L359-L368].

74515876. Cannot find module '/workspace/server.js' on Google Cloud (App Engine)

Asked Nov 21, 2022 by David (Score: 0, Tags: deployment, gcloud, google-cloud-deploy).

Problem: Deploying a Node.js application to Google (App Engine Standard). Build failed with “Cannot find module '/workspace/server.js'”. The app’s code was compiled into a build/ folder. The user suspected missing package.json or entrypoint.

No Answer: The question has **0 answers** visible (answer content likely deleted). Many comments discussed package.json and app.yaml. Without an accepted answer, this is unsolved in posts. Likely fixes: ensure package.json is deployed or an entrypoint is set in app.yaml.

Root Cause (inferred): On GAE Standard, the main file (like server.js) was not found. Possibly missing package.json or build output mis-specification.

Solution (external): For App Engine Standard Node.js: either include package.json and have start script, or specify entrypoint: node build/index.js in app.yaml. Also ensure npm

run build outputs to ./ if needed.

No official answer to cite. We note unresolved question.

72293744. GCP deploy error: Creating Revision interrupted

Asked May 18, 2022 by Adam (Score: 3, Tags: google-cloud-run google-cloud-build terraform-cloud google-cloud-deploy).

Problem: Cloud Build step using gcloud beta run services update consistently failed with “*Creating Revision.....interrupted*”. Build logs show “*ERROR: (gcloud.beta.run.services.update) Revision <name> is not ready.*”. Possibly due to specifying --tag or --labels.

GCP Services: Cloud Run, Cloud Build, Terraform.

Accepted Answer (Christian, score 2): The problem was due to assigning a **traffic tag** to a non-ready revision. When a revision has 0% traffic but is not yet ready, the update fails. The fix is to **remove any traffic assignments** from that revision. In practice: unassign the tag via GCP Console or CLI (remove --tag or set --traffic appropriately) and retry.

Other Answer (Andrés, score 0): Suggested removing revision/tag from YAML; pointed out that a “0% traffic” assignment still causes the error.

Root Cause: Assigning traffic (even 0%) to an unready revision.

Solution Steps:

1. In your deployment YAML or Cloud Build script, **remove the --tag (revision) or --traffic settings** that target the new revision. Let Cloud Run assign traffic after the update.
2. Via GCP Console or gcloud, clear any traffic targets on non-ready revisions.
3. Rerun the deployment. It should complete successfully.

No code snippet needed. See description. Affected services: Cloud Run, Cloud Build.

Links: *Cloud Run revision traffic docs (implied).*

70946280. Cloud Function deploy fails due to “--production=false”

Asked Feb 1, 2022 by Zachary Johnson (Score: 3, Tags: google-cloud-platform google-cloud-functions gcloud google-cloud-deploy).

Problem: Deploying an existing Cloud Function (Node.js 16) began failing with: “*Invalid option name ('--production=false')*”. The user was not passing --production=false. The logs showed: “*Manifest is still referenced by tag: latest*”, indicating a build failure with Yarn v2/3.

Accepted Answer (Zachary Johnson, OP, score 1): This was a Google-side bug with the buildpack and Yarn 2/3. Google released a fix (shortly after the issue). After the fix, deploys work again.

Root Cause: A regression in Google’s Node.js buildpacks causing Yarn to inject --

production=false.

Solution: No action needed now – update to latest gcloud or wait for Google’s fix. In practice, the fix was deployed (comment notes it’s working again). Alternatively, downgrade Yarn to v1 as workaround.

Link: Issue on GitHub (GoogleCloudPlatform/buildpacks).

69829161. Use Cloud Build privatePool in Cloud Deploy

Asked Nov 3, 2021 by insidehustle (Score: 0, Tags: google-cloud-build gcloud-deploy).

Problem: The user has a shared VPC with a GKE cluster in one project and Cloud Deploy pipeline in another. They defined a Cloud Deploy target with executionConfigs.privatePool.workerPool pointing to another project’s Cloud Build private pool, but deployments hung (no build triggered in either project). They gave all possible IAM roles. Cloud Deploy still built in its own project, not the private pool project.

Accepted Answer (Edward Thiele, score 0): Cloud Deploy **always runs the build in the same project as the pipeline**, even if you specify a private pool in another project.

However, you *can* use a worker pool from another project:

1. Grant cross-project permission to use the pool (see Cloud Deploy docs).
2. Ensure the Cloud Deploy service account has logging permission on the deploy project (for logs to appear).
3. Grant Cloud Deploy “act-as” permission if the build’s service account is from a different project.

In short, Cloud Deploy will run in the Cloud Deploy project but can leverage a foreign pool, with proper IAM setup.

Root Cause: Cloud Deploy’s design (build execution project fixed) and insufficient cross-project IAM.

Solution Steps:

4. Follow [Cloud Deploy execution environment docs](#) to allow using a private pool across projects (set up IAM in the pool’s project).
5. In Cloud Deploy target YAML, specify executionConfigs.privatePool.workerPool (even in other project).
6. Ensure Cloud Deploy’s service account has logs.writer on the deploy project for log visibility, and is permitted to *act as* the Cloud Build service account used in the private pool.
7. Trigger a release; logs should now appear in the Cloud Deploy project’s logs.

Links: Execution environment for private pools docs.

Issue Categories & Affected Products

The table below summarizes the frequency of each root-cause category across the 128 questions:

Category	Count
Syntax/Config errors (YAML/Helm)	35
CLI/Flags/Command misuse	22
Scaffold/pipeline configuration errors	18
Cloud Deploy design/limitations	15
Networking/permissions (private GKE)	12
Runtime/version mismatch (AppEngine)	8
Terraform conflict	6
Buildpack/tool bugs (e.g. Yarn)	3
Firebase vs gcloudignore	1
Others/Unresolved	8

The most common affected GCP services (from tags) were:

GCP Service/Product	Mention Count
Cloud Deploy	128†
Cloud Run	10
Cloud Build	9
Scaffold	7
GKE/Kubernetes Engine	6
Cloud Run (Knative)	5
App Engine (GAE)	4
Helm (charts)	3
Cloud Functions	2
Deployment Manager (DM)	2
Cloud Build Private Pool	2
Firebase Hosting	1

(Every question includes Cloud Deploy tag; counts above from additional service tags.)

The table of common solution patterns shows the counts of identified fixes:

Solution Pattern	Count
Correct syntax/indentation (YAML/Helm)	33
Add missing CLI flag or parameter (gcloud, skaffold)	20
Use deploy parameters or separate pipelines/profiles	16
Use proper Skaffold configuration (profiles, manifests)	15
Use gcloud run services update --image=... or similar	6
Configure execution environment (private pool)	5
Fix networking/VPN/BGP config	4
Update runtimes (nodejs18) or libraries	4
Provide entrypoint (function/handler)	3
Wait for upstream fix or workaround (e.g. Yarn regression)	2

(Counts include answers, not all questions, and some overlap.)

Decision Flows for Top Issue Types

mermaid

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flowchart TD

A[Start: Deployment fails] --> B{Error is syntax/param in YAML?}

B -->|Yes| C[Check for missing colons, indentations, required fields]

C --> D[Apply fix: correct YAML (from SO examples) **【45†L194-L201】** **【96†L161-L168】**]

D --> E[Re-deploy]

B -->|No| F{Error in CLI/flags usage?}

F -->|Yes| G[Review `gcloud` command flags]

G --> H[Add missing flag (e.g. --function, --skaffold-file) or fix parameters **【12†L231-L239】** **【16†L149-L158】**]

H --> E

F -->|No| I{Using Skaffold/CloudDeploy profiles?}

I -->|Yes| J[Inspect `skaffold.yaml`]]

J --> K[Ensure `deploy` block is under profile and rawYAML lists all services 【59†L228-L236】 【91†L349-L357】]

K --> E

I -->|No| L{Mixing target types?}

L -->|Yes| M[Cloud Deploy supports one target type per pipeline 【20†L202-L210】]

M --> N[Use separate pipelines or deployParameters per target 【20†L202-L210】 【57†L232-L239】]

N --> E

L -->|No| O{Network access issue?}

O -->|Yes| P[Configure execution environment/private pools]

P --> Q[Enable private pool and VPC peering; fix VPN/BGP config 【79†L143-L151】 【83†L359-L368】]

Q --> E

O -->|No| R[Other issues: version mismatch or bug]

R --> S{App Engine on Node14?}

S -->|Yes| T[Update runtime or libraries 【55†L163-L171】 【36†L85-L94】]

S -->|No| U{Cloud Function Yarn bug?}

U -->|Yes| V[Wait for GCP fix (Yarn regression) 【104†L205-L213】]

U -->|No| W[Investigate further or ask new Q]

Open Questions / Limitations

- **Unanswered Questions:** A few questions had no accepted answers (e.g. the App Engine “server.js” issue). Our report notes them but solutions are inferred.
- **Potential Gaps:** We rely on answers with score ≥ 0 . Some low-vote answers or deleted answers may contain relevant info not captured. Wherever no accepted answer, we mention known best practices or mark as unresolved.

- **Classification Ambiguity:** Some issues span categories (e.g. Cloud Run config vs Terraform conflict). We assign primary cause as above, but overlap exists.
- **Rate Limits:** We processed all available questions. No items were skipped for rate limits.